

Elisa M. Varghese

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EDUCATION

University of California, San Diego

B.S. in Data Science

American High School

Diploma

- **GPA:** 4.36/4.00, *Academic Block Award*
- **Relevant Coursework:** Intro into Computer Science, AP Computer Science Principles, AP Computer Science A, AP Statistics, AP Calculus AB

La Jolla, California

Expected Graduation, June 2029

Fremont, California

2021-2025

EXPERIENCE

Support Vectors

Data Science and AI Intern

- Built NLP pipelines with Python + spaCy to analyze classical literature
- Designed RAG apps with OpenAI API to generate slogans and AI-assisted business plans
- Produced a 40+ page AI-driven business plan integrating LLM outputs

Fremont, California

July 2024 – Sep 2024

Malayalam Class

Director and Lead Instructor

- Shortlisted and cold-called 300+ potential attendees, created 30+ slideshows, graded 900+ assignments with individualized feedback, and designed 90+ certificates
- Delegated roles during frequent meetings with co-teachers and guided students from ages 4-53

Fremont, California

June 2021 – July 2025

Habitat For Humanity

President, Secretary, Underclassman Representative, Publicist

- Led 35+ volunteers in community service events, including trash pickups, fundraisers, and construction projects.
- Partnered with small businesses to raise funds for charitable causes, oversaw club communications, and designed promotional materials

Fremont, California

Aug. 2021 – May 2025

PROJECTS

Literature NLP Analyzer

Internship Project

- Designed an NLP pipeline in Python using spaCy to tokenize, extract, and rank character names across multiple classical texts
- Implemented text preprocessing (stopword removal, lemmatization) and frequency analysis to identify key patterns in character mentions
- Visualized results with Pandas, producing frequency charts that revealed structural trends in narrative focus

AI Business Plan Generator

Internship Project

- Built two Retrieval-Augmented Generation (RAG) applications using OpenAI API and prompt engineering to generate creative marketing slogans and structured business content
- Led the development of a pipeline that integrated LLM outputs into a cohesive 40+ page AI-assisted business plan
- Optimized prompt templates to balance clarity, creativity, and factual accuracy, resulting in slogans that scored highly in evaluation

Flag Guessing Game

AP CSP Project

- Programmed an interactive flag-guessing game in Python, implementing multi-level difficulty (Levels 1–3)
- Designed scoring logic where higher-level points were earned only after completing prior levels, with bonus games awarding triple score multipliers
- Applied control structures, conditional logic, and data structures to track player progress and calculate cumulative scores
- Enhanced user experience with randomized questions and replay functionality, creating a replayable educational game

AWARDS AND HONORS

- Richard D. King Rotary Club Speech Contest Finalist (District 2024, Regional 2025).
 - Advanced through multiple levels by earning first place or top two; first person from the Fremont Rotary Club to make it to the district finals in the 23 years since the competition's inception and was 1 of 2 finalists
- PVSA Award (Gold 2023, Bronze 2024).
 - Awarded to teenagers who have completed at least 100 hours of volunteer service under a verified organization within a one-year period; first recipient of the award under the California Malayali Youth League and also went on to earn it a second time.
- National Mar Gregorios Syriac Orthodox Students Association 10th Grade Exam – Distinction (2023).
 - Awarded to students who pass the exam with 90% or above; first recipient in my church's history, the only recipient in my diocese's Western Region, and among ~25 recipients nationwide in 2023.

SKILLS

Programming: Python, Java, C++, Pandas, spaCy, OpenAI API, Git/GitHub

Data: NLP, RAG, Visualization, Prompt Engineering

Languages: English, Malayalam, Spanish, Syriac